Deep Research Prompt: Rewriting HTNLA as a Rigorous Nexus v4 Subsystem Paper

How to use this: Paste this whole prompt into a frontier reasoning model with real web search / deep research capability. Attach or paste immediately after it: the current HTNLA draft ("Hypervector Nonlinear Transdimensional Algebra (HTNLA): A Unified, Lossless Blueprint," including its verbatim Research-dossier archive). This prompt is the companion to the HTLA deep-research prompt already produced — HTNLA is the nonlinear extension of that linear substrate, and the two rewrites must agree with each other wherever they overlap.

Mission

HTNLA is registered in the Nexus 4 corpus as **S9, a Tier 2 subsystem-math source**. The master monograph's Part III already contains a short, correctly-hedged distillation of it (§6, quoted in full below) — component-wise nonlinear activation on top of HTLA's block operators, an explicit five-way taxonomy for degrees of invertibility loss, no invented benchmark numbers, no consciousness-metric cosplay. The current standalone draft is not that. It has real bones (per-label nonlinearity, k-WTA sparsity, residual connections are all legitimate, useful ideas) buried under invented statistics, at least two pieces of code that don't run, and — the big one — several thousand words of a *different document* (a human identity-transfer research protocol) stapled onto the back of a linear algebra paper.

The job: produce the long-form, fully rigorous, honestly-scoped version of HTNLA that the master monograph's condensed §6 can safely point to, capable of replacing S9 outright.

Ground truth you must not contradict (Nexus v4 canon, current as of the 2026-08-28/31 monograph)

§6 of the master monograph, verbatim — this is the spine the rewrite must elaborate on, not deviate from:

HTNLA extends the linear block calculus by inserting nonlinear maps while preserving label boundaries. A generic target component is z_t = b_t + Σ_s W_{t,s}x_s + r_t, followed by y_t = φ_t(z_t), and, where configured, y_t ← kWTA_t(y_t). A multi-layer nonlinear transdimensional network may be written H^(k+1) = σ^(k)(M^(k)H^(k) + b^(k)), optionally with a residual map when source and target label geometry permits: H^(k+1) = R^(k)H^(k) + σ^(k)(M^(k)H^(k) + b^(k)).
6.1 What nonlinearity changes. Linear invertibility cannot be assumed after a non-injective nonlinearity. If φ collapses distinct states, $\varphi(x_1)=\varphi(x_2)$, $x_1 \neq x_2$, then an exact inverse does not exist. HTNLA must therefore distinguish: exact inverse where the operator is bijective on the relevant domain; pseudoinverse where a

least-squares approximation is defined; partial unbinding where only some coordinates are recoverable; iterative/resonator reconstruction where the inverse is a search problem; unknown or ambiguous outputs where information has genuinely been lost. This distinction is a recurring Nexus principle: uncertainty is represented, not papered over.

The reference geometry this must stay consistent with: the quinary binding operation is a deterministic, deliberately information-losing intersection-plus-sign-composition-plus-conservative-magnitude operation (not circular-convolution-plus-tanh), with a four-state partial-unbind taxonomy: `EXACT / AMBIGUOUS / UNKNOWN / IMPOSSIBLE`. HTNLA's own five-way invertibility taxonomy above must map onto this, not reintroduce a separate, less rigorous scheme.

§73 of the master monograph ("Consciousness Claims and Category Errors") governs every biologically- or consciousness-flavored name in this draft (dopamine, serotonin, GABA, Hebbian/STDP, "phenomenal continuity," Φ, "cognitive fidelity"). It distinguishes three levels — computational variable / functional analogue / phenomenological claim — and states plainly: "evidence for level 1 does not establish level 3." Apply this explicitly to every such term in the rewrite rather than letting the borrowed vocabulary imply more than it's earned.

Also standing: the 12-term claim-status vocabulary (`CANONICAL, IMPLEMENTED, VERIFIED, PARTIALLY IMPLEMENTED, ACTIVE DEVELOPMENT, PROPOSED, DESIGN TARGET, EXPERIMENTAL, RESEARCH HYPOTHESIS, APPLICATION-SPECIFIC, HISTORICAL/SUPERSEDED, INFERRED`) — reuse it, don't invent a new one. And the already-ratified correction on the "85% energy conservation" number: it is a design motif, not a measured law, and any energy claim needs real telemetry, not a percentage asserted as fact.

Major structural issue: this document is two documents, and only one of them is HTNLA

Sections 6 and 9.6 through 9.12.26 of the draft (Phenomenal Continuity and Identity Preservation, the Hybrid Identity Transfer Protocol, QHRF measurement — Brillouin microscopy, NV-diamond magnetometry, specific laser wattages and pulse widths — PyMC hierarchical fidelity models, NWB/JSON schemas for a human-subjects project, and a literal IRB/consent checklist including "destructive transfer") have essentially nothing to do with nonlinear block operators, activation functions, or k-WTA sparsity. They belong to a different, adjacent question — what does substrate/identity continuity mean — that Nexus v4 has **already answered better elsewhere**: the current Identity Architecture canonical monograph grounds continuity in AmmunIty (a security/homeostasis application) and the Educated Bookseller business lineage, not in neural-signal capture instrumentation. Keeping the QHRF material inside the HTNLA math paper does three bad things at once: it makes a legitimate linear-algebra extension look like it's smuggling in a brain-uploading proposal; it duplicates and contradicts a concern the corpus has already resolved more soberly; and it guarantees that any technically literate reader who gets past the math will hit the identity-transfer material and stop trusting the rest of the document.

Required action: remove sections 6 and 9.6–9.12.26 from the HTNLA paper entirely. If this material has ongoing value as TUCR's speculative/fictional layer, it belongs in its own separate, explicitly-labeled document (worldbuilding, not a research proposal) — not stitched into either the math papers or the real Identity Architecture engineering monograph. Do not summarize it, footnote it, or preserve a "lossless" pointer to it inside the HTNLA rewrite; a math paper does not need a cross-reference to a human-subjects protocol.

Defects in the retained (actual HTNLA) material — fix all of these

#	Location	What's wrong	Fix
1	§3.3, "graded $\mathbb{R}$-algebra" → "graded nonlinear algebra"	Once nonlinearity enters, this isn't just imprecise the way it was for HTLA — it stops making formal sense at all. Two nonlinear layers $\sigma(M_1H)$ and $\sigma(M_2H)$ don't even satisfy $\sigma(M_1H)+\sigma(M_2H) = \sigma((M_1+M_2)H)$ in general, so the space of HTNLA layers isn't closed under addition the way a linear operator space is — there is no algebra here to be "graded."	Reframe: HTNLA defines a family of parameterized nonlinear maps built on top of HTLA's block structure (which *is* correctly framed via the Peirce-decomposition/ringoid treatment from the HTLA rewrite). Call it a network/architecture family, or a category of labeled nonlinear morphisms — not an algebra, graded or otherwise.
2	§3.3, tanh-based nonlinear binding, `bind(H,K) = tanh(H⊙K+b)`	Diagnosed as causing "non-invertible in a direct sense" — but tanh is actually invertible on its open range via artanh. The real sources of information loss are saturation/precision near ±1, the unknown/untracked bias b, and the pre-existing approximate nature of circular-convolution binding itself (already true before any nonlinearity is added). This entire construction is also inconsistent with the canonical quinary binding	Remove tanh-based binding. Describe HTNLA's binding contribution as: how (if at all) a differentiable/soft nonlinear analogue is used during *training* of learned components, explicitly distinct from and non-authoritative over the deterministic quinary reference binding — and map the five-way invertibility taxonomy above onto the quinary system's EXACT/AMBIGUOUS/UNKNOWN/IMPOSSIBLE states rather than inventing a parallel one.

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		operation, which was adopted specifically to replace it.	
3	§3.4, §5.1, §6.2 (three separate occurrences), "70% sparsity constraint reduces the computational cost from O(D²) to O(D log D)"	Identical error to the one already corrected in the HTLA rewrite: a fixed retention fraction gives a constant-factor speedup (~3x at 70% sparsity), not a complexity-class change. Real O(D log D) comes from FFT-based structure or from block-level coupling-graph sparsity that shrinks relative to D, not from k-WTA on a dense vector.	Apply the identical fix specified in the HTLA prompt. Do not re-derive it differently here — the underlying mathematical mistake is the same regardless of which paper it appears in.
4	Abstract vs. §1.2	Abstract states as achieved fact: "achieving 20-50% improvement in associative recall accuracy." §1.2 says "Projections *indicate* a substantial 20-50% improvement." Same number, hedge present in one place and silently dropped in the other — inside a single document. No experiment anywhere backs either version.	Drop the number entirely unless a real ablation is run. If a directional claim is wanted, state it as a single, consistently-hedged `RESEARCH HYPOTHESIS` with a stated test protocol (see the ablation checklist already sketched in §9.4/9.12 of the draft — that part is a legitimate, keepable idea, it's the pre-loaded "answer" that isn't).
5	§6.2, §7.1/8.2 tables — Φ, Binding Score, Neural Dynamics Fidelity, Memory Accuracy, Hilbert Distance	Precise numeric thresholds (Φ<0.3, +15%, Binding Score>0.8) are set for metrics with **no formula given anywhere** for how any of them is actually computed from HTNLA state. This is exactly the pattern §73 exists to prevent.	Either drop these "cognitive fidelity" metrics from a math/architecture paper entirely, or give each one an explicit level-1 computational definition (a real formula) and tag it accordingly — no numeric threshold survives without one.

#	Location	What's wrong	Fix
6	Nearly every substantive claim in the document, including the most specific ones (20-50%, 85%, O(D log D), Φ<0.3)	All trace to a single repeatedly-cited internal source ("HVD Architectural Plan - The TMS HDF5 Hierarchical Model.txt," footnoted "1" dozens of times) with no independent external verification anywhere. Citing the same unverified internal document forty times is not forty pieces of evidence — it's one unverified claim, repeated.	Anything above `RESEARCH HYPOTHESIS` needs either a real external citation or a derivation/execution the reader can check themselves.
7	§7.3/8.3 comparison table, "Expressive Power: High (Universal Approximator)"	True for some member of this architecture family at unbounded width; not established for the specific fixed-sparsity, few-layer network actually proposed.	Cite the universal approximation theorem's actual hypotheses (width/depth conditions) and state plainly whether the proposed architecture is known to satisfy them, or soften the cell.
8	§9.9, neuromodulatory formulas (serotonin-modulated attention, GABA sparse-coding constraint, dopamine-modulated Hebbian rule) with specific unexplained constants	Real neuroscience vocabulary attached to invented functional forms and constants (γ, τ_focus, σ_att, θ_inhib) with no derivation or citation.	See Open Research Questions #3 below — either ground these in the real reward-modulated-STDP literature with correctly attributed functional forms, or explicitly tag as `INFERRED`/loosely-inspired-by, never as if the specific constants came from somewhere.
9	Phased roadmap (§1.2, and the verbatim archive's §6.1 "Phase 1 Months 1-3... benchmarked...	Project planning dressed with the same false precision as the math.	Tag with `PROPOSED` / `DESIGN TARGET` per the existing vocabulary; this is a minor, mechanical fix once the tagging discipline is applied consistently.

#	Location	What's wrong	Fix

target latency
<10ms per layer")

Verified code bugs — confirmed by execution, not inspection

`risk_aware_modulation`: as literally written, `np.dot(M, H) + 0.1*DA*H` is a shape-mismatch crash the moment M's output dimension differs from H's input dimension (confirmed: `ValueError: operands could not be broadcast together with shapes (50,) (100,)` on the paper's own example dimensions). Separately, and more importantly: `risk = np.var([reward] * 10)` computes the variance of ten identical copies of one number, which is **mathematically always exactly 0**, for any reward value — confirmed by execution. The entire "risk-aware" temperature modulation this feeds is inert by construction; whatever real signal was intended for `risk`, this formula cannot carry it. Needs a real variance computation over an actual distribution of outcomes (e.g., a rolling window of past rewards) if the risk-modulation idea is worth keeping.
`dynamic_label_adapt`: `current_mean = np.mean([hypervector.components[l] for l in labels], axis=0)` requires every label's component to have the same dimension to stack — confirmed to throw a numpy inhomogeneous-shape error the moment two labels have genuinely different dimensions, which is the entire premise HTLA/HTNLA exist to handle. Needs to either operate per-label or use an explicit padding/projection step before any cross-label averaging, with the dimension-adaptation logic actually tested against heterogeneous inputs before publication.
General instruction: every code listing in the rewrite must be executed against inputs with genuinely different per-label dimensions (not all-same-size toy examples) before being included, since that's precisely the condition under which several of these functions break.

Open research questions — actually investigate these

Novelty scoping for the nonlinear extension specifically. Beyond the HTLA-level prior art already identified (Named Tensor Notation, categorical deep learning), HTNLA's actual closest relatives are: Graph Neural Networks with per-edge-type/per-node-type transformations (structurally very close to per-label blocks with per-label nonlinearities); Mixture-of-Experts architectures (sparse routing closely analogous to k-WTA over labels); modular/multi-task networks with task-specific heads (the "Splitting" operator concept). Survey these and state honestly what HTNLA's label-plus-epistemic-status combination adds that they don't already do.
Invertibility taxonomy grounding. The "exact / pseudoinverse / partial unbinding / iterative reconstruction / unknown" five-way distinction should be connected to the real, well-established Invertible Neural Networks / Normalizing Flows literature (e.g., RealNVP, Glow, i-RevNet) — this is exactly the right existing research area for reasoning rigorously about designing neural network layers with controlled, provable invertibility properties, and is far stronger grounding than "gradient ascent" hand-waving.

Dopamine-modulated learning grounding. Real, correctly-attributable literature exists: Izhikevich (2007, *Cerebral Cortex*, "Solving the distal reward problem through linkage of STDP and dopamine signaling"), and reward-modulated-STDP theory in Frémaux, Sprekeler & Gerstner (2010, *J. Neuroscience*) and Legenstein, Pecevski & Maass (2008, *PLOS Comp. Biol.*). Determine whether the draft's specific functional forms can be honestly attributed to (a correctly-cited, explicitly-labeled functional analogue of) this literature, or whether they should be simplified and clearly marked `INFERRED`.
k-WTA's actual stabilizing/concentration properties. The draft asserts sparsity "acts as a stabilizing force" and "preserves the Concentration of Measure property" without derivation. Search sparse-coding and winner-take-all computational literature (e.g., work descending from classical WTA circuit theory, and sparse-distributed-representation literature) for any result that would actually support this, and either cite it properly or downgrade the claim.
Quantum nonlinearity — this is a known hard problem, not a solved primitive. The draft's claim that "nonlinear transformation F can be block-encoded into a unitary matrix U_F" glosses over a real, actively-published-on difficulty: quantum evolution is fundamentally linear/unitary, and implementing genuine nonlinearity requires ancilla qubits with measurement-induced (probabilistic) collapse, repeat-until-success protocols, or hybrid quantum-classical splits — confirmed as an open, non-trivial research area across multiple 2020–2025 papers on quantum neural networks. Rewrite this section to state the real difficulty honestly, citing the actual quantum-neural-network literature on nonlinearity implementation, rather than treating block-encoding of a nonlinear map as a routine extension of the (genuinely more tractable) linear QSVT machinery used in the HTLA quantum appendix.

What "legitimate" and "novel" mean here

Same standard as the HTLA rewrite: label every claim at the confidence it has earned, and locate the genuine novelty in the actual combination, not in individual borrowed tools. For HTNLA specifically, the defensible thesis is: a component-wise nonlinear extension of a labeled block-linear substrate, paired with an explicit, typed taxonomy for degrees of invertibility loss (rather than silently assuming or silently discarding invertibility), integrated with a concrete deterministic reference implementation. That is a real, useful, buildable contribution. "Transforming a memory substrate into a fully adaptive, self-optimizing cognitive engine" suitable for "nano-LLMs" is not a claim this paper can support and shouldn't attempt to.

Required output structure

Title + abstract calibrated to what's proven, cited, or explicitly hypothesized. No "foundational," "profound," "self-optimizing cognitive engine," or consciousness-flavored language in the framing.
Short "Relationship to Prior Art" section (GNNs/MoE, invertible NNs, k-WTA/sparse coding, reward-modulated STDP if kept).
Core mathematics: the layer definition and multi-layer/residual form exactly as already canonized in §6 above; the five-way invertibility taxonomy mapped explicitly onto the quinary

EXACT/AMBIGUOUS/UNKNOWN/IMPOSSIBLE states.
Sparsity (k-WTA): correctly scoped complexity claims (constant-factor, not complexity-class, unless block-sparsity is doing the real work).
Any biologically-named mechanism kept: explicitly tagged per §73's three levels, with real citations where claimed.
Reference implementation: real, executed code only, tested against heterogeneous per-label dimensions.
Closing claim-status audit table, reusing the 12-term vocabulary.
No section 6-equivalent or 9.6-9.12-equivalent material — phenomenal continuity, identity transfer, and QHRF instrumentation do not appear in this document at all.

Hard constraints

No unqualified superlatives, no consciousness-science vocabulary used without the §73 three-level tagging applied.
No number without a citation or a pointer to code that was actually run to produce it.
No claim cites the single internal architecture-plan document as if repetition were verification.
Every equation is proved, cited, or explicitly tagged `RESEARCH HYPOTHESIS` / `INFERRED` — no fourth option.
The identity-transfer/QHRF material does not appear anywhere in this document, including as a summary, a footnote, or a "see also" pointer.